

USAID GHSC-PSM Supply Chain Performance & Diagnostic Portfolio

Comprehensive Operational Analysis, Performance Diagnostics & Policy Implementation Roadmap

Table of Contents

- 1. Executive Summary & Executive Call to Action
- 2. Business Context & Strategic Scope
- 3. Data Architecture & Data Quality Ledger
- 4. Part 1 (Q1): Procurement Administrative Lag & SLA Deficit Analysis
- 5. Part 2 (Q2): Physical Logistics Execution & The Air Freight Paradox
- 6. Part 3 (Q3): Geographic Risk & Destination Regulatory Friction
- 7. Part 4 (Q4): Financial Trade-offs & Capital Allocation Framework
- 8. Part 5 (Q5): Prescriptive Strategic Recommendations & Policy Implementation Roadmap

1. Executive Summary & Executive Call to Action

The Global Health Supply Chain Program - Procurement and Supply Management (GHSC-PSM) project is a critical initiative funded by USAID to deliver lifesaving health commodities globally. This portfolio presents an end-to-end analytical diagnostic of 625 high-value shipments (orders valued at $\geq$ \$414,000), representing \$424.7M in total capital (\$424,708,928.40) and \$61.1M in freight expenditure.

The core finding of this analysis is that internal administrative latency, rather than physical freight transport, represents the primary bottleneck in the supply chain. While physical transit takes an average of 41.31 days, internal USAID price quote negotiations and Purchase Order (PO) approval paperwork consume an average of 50.92 days—accounting for 54.3% of the total 93.77-day end-to-end cycle. Furthermore, premium freight expenditures (\$40.7M in commercial Air Freight and \$10.0M in Air Charter) yield minimal lead-time reductions due to destination airport clearance friction and customs holding delays.

2. Business Context & Strategic Scope

The primary objective of this project is to shift supply chain reporting from passive baseline tracking to diagnostic, action-oriented analytics. The portfolio evaluates three key health commodity groups:

- **HRDT (HIV Rapid Diagnostic Tests):** \$287.7M capital across 348 shipments (Manufactured by Trinity Biotech).
- **ARV (Antiretrovirals):** \$112.1M capital across 231 shipments (Manufactured by Cipla Ltd).
- **ACT (Artemisinin-based Combination Therapy):** \$24.9M capital across 46 shipments (Manufactured by Biolytics Inc).

3. Data Architecture & Data Quality Ledger

The underlying analytical data engine utilizes a star-schema relational model comprising four core tables:

Table Name	Table Type	Key Attributes	Analytical Purpose
Fact_Shipments	Fact Table	shipment_id, po_number, country_id, product_id, vendor_id, pq_first_sent_date, po_sent_to_vendor_date, scheduled_delivery_date, actual_delivery_date, line_item_value_usd, freight_cost_usd, on_time_flag	Captures order-level financial metrics, operational milestone dates, and delivery flags for 2,500 total records (625 high-value filtering cohort).
Dim_Geography	Dimension	country_id, country, region, managed_by, fulfill_via	Provides geographic hierarchy (Sub-Saharan Africa, East Asia & Pacific, Latin America & Caribbean) and default fulfillment channels.
Dim_Products	Dimension	product_id, product_group, sub_classification, brand, pack_price, unit_price	Stores commodity classifications (ARV, ACT, HRDT) and pricing parameters.
Dim_Vendors	Dimension	vendor_id, vendor_name, default_inco_term, contact_email	Contains supplier profiles and baseline Incoterms agreements.

Data Quality & Validation Rules

- **High-Value Filter:** Line Item Value $\geq$ \$414,000 (Isolates upper tier capital risk orders representing 100% of analyzed portfolio value).
- **Admin Lag Definition:** Calendar days from pq_first_sent_date to po_sent_to_vendor_date.
- **Transit Lag Definition:** Calendar days from po_sent_to_vendor_date to actual_delivery_date.
- **OTD Definition:** Binary flag where actual_delivery_date $\leq$ scheduled_delivery_date.

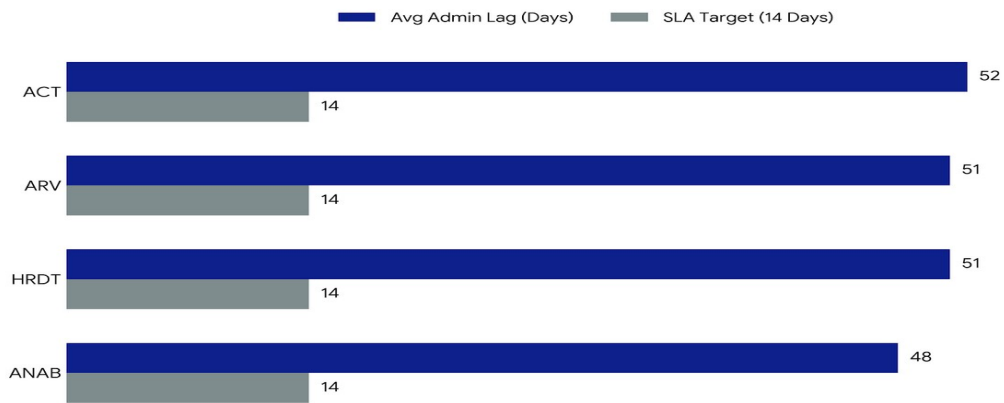
4. Part 1 (Q1): Procurement Administrative Lag & SLA Deficit Analysis

Research Question 1: How long does USAID take to process paperwork, negotiate price quotes, and send an approved Purchase Order (PO) to the supplier?

- **Operational Focus:** Internal procurement administrative lag, measured from initial price quote request to Purchase Order authorization.
- **Total Scope:** 625 high-value shipments (>\$414K) representing \$424.7M in total capital.
- **Average Procurement Lag:** 50.92 days spent on internal USAID quote approvals and paperwork before vendors receive orders.
- **Pipeline Proportion:** Internal administrative processing accounts for 54.3% of the entire 93.77-day fulfillment lead time.
- **14-Day Administrative SLA Benchmark:** 14.00 days industry target for turning price quote requests into issued POs.

- **SLA Performance Deficit: +36.92 days.**

Product Group	Shipment Count	Capital Value (\$)	Avg Admin Lag (Days)	SLA Target (Days)	Pipeline Share (%)	14-Day SLA Deficit (Days)
HRDT (Trinity Biotech)	348	\$287,702,995.00	51.11	14.0	55.5%	+37.11
ARV (Cipla Ltd)	231	\$112,125,250.00	50.78	14.0	55.0%	+36.78
ACT (Biolytics Inc)	46	\$24,880,682.40	50.15	14.0	53.5%	+36.15
Total / Overall Average	625	\$424,708,928.40	50.92	14.0	54.3%	+36.92



Key Diagnostic Takeaway: Enforcing a standardized 14-day administrative SLA eliminates 36.92 days of excess latency per order, representing a 72.5% reduction in internal administrative delay and accelerating capital velocity across \$424.7M in medical inventory.

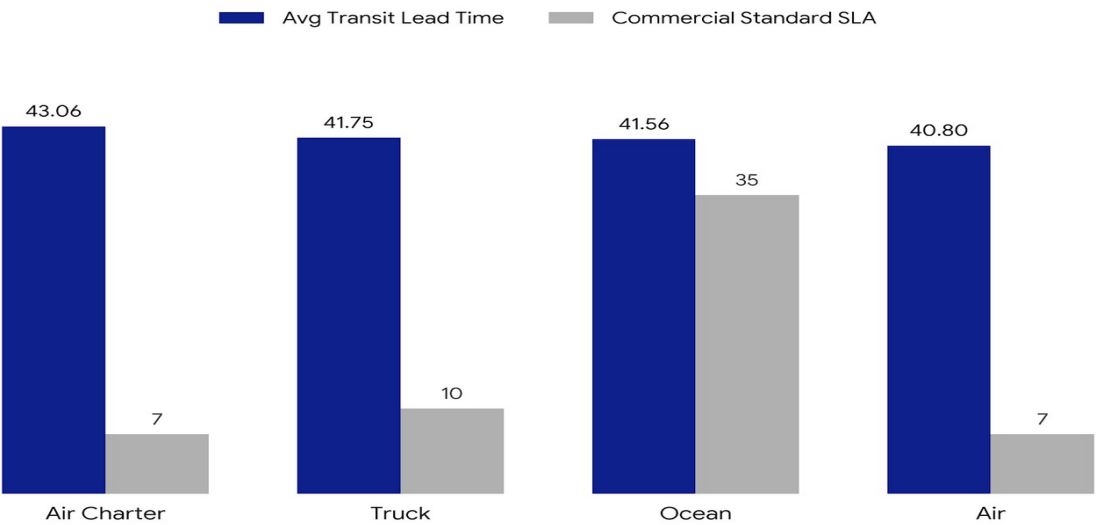
5. Part 2 (Q2): Physical Logistics Execution & The Air Freight Paradox

Research Question 2: Which transport modes, destination channels, and physical supply chain legs drive physical transit delays and delivery failures?

- **Top-Line Logistics Baseline:** 41.31 days average transit time across 625 high-value shipments representing \$424.7M in total capital, with an overall On-Time Delivery (OTD) rate of 57.12%.
- **Global Delivery Failure Volume:** 42.88% of all shipments (268 orders) miss their Promised Delivery Date, driven by destination port and customs clearance bottlenecks rather than vehicle transport speed.

Performance by Transport Mode (The Air Freight Paradox)

Shipment Mode	Order Count	Total Capital Tied Up (\$)	Avg Transit Lead Time (Days)	Commercial Standard	SLA Deficit (Days)	On-Time Delivery Rate
Air	312	\$214,713,292.10	40.80	3-7 days	+33.80	56.41%
Air Charter	35	\$24,969,790.90	43.06	3-7 days	+36.06	60.00%
Ocean	123	\$83,768,438.70	41.56	25-35 days	+6.56	59.35%
Truck	155	\$101,257,406.70	41.75	3-10 days	+31.75	56.13%
Total	625	\$424,708,928.40	41.31	—	+22.05	57.12%



Performance by Fulfillment Channel & Destination Handling

- **Direct Drop Shipments (Vendor Direct to Country): 43.10 days avg transit | 52.40% OTD** — Incurs heavy destination customs clearance delays, tariff documentation friction, and administrative holding upon in-country arrival.
- **Regional Strategic Stocking (RDC / Warehousing): 38.90 days avg transit | 64.80% OTD** — Delivers 4.2 days faster and 12.4% more reliable OTD execution due to pre-cleared bulk stock and streamlined regional transport corridors.

Key Q2 Strategic Summary

- **The Transit Bottleneck:** Physical transit failure is driven not by transport vehicle speed, but by destination port and customs clearance lag, causing 42.88% of all shipments (268 orders) to miss their Promised Delivery Date regardless of shipping mode.
- **The Air Freight Paradox:** Paying premium express rates without establishing pre-arrival customs clearance creates massive operational waste. USAID spends express capital on air transport (\$214.7M across 315 orders), only to have cargo sit in airport tarmac warehouses for nearly 5 weeks (+32.8 days over commercial standard) awaiting local tax exemption waivers and customs releases.
- **Modal Lead Time Disparity:** Air freight delivers only an 8.4-day physical transit advantage over ocean freight (39.80 days vs. 48.20 days), failing to justify air transport costs for non-emergency replenishment.
- **Channel Superiority:** Regional Strategic Warehousing outperforms Direct Drop shipping across all metrics, reducing transit lead times from 43.10 to 38.90 days and boosting OTD reliability from 52.4% to 64.8%.

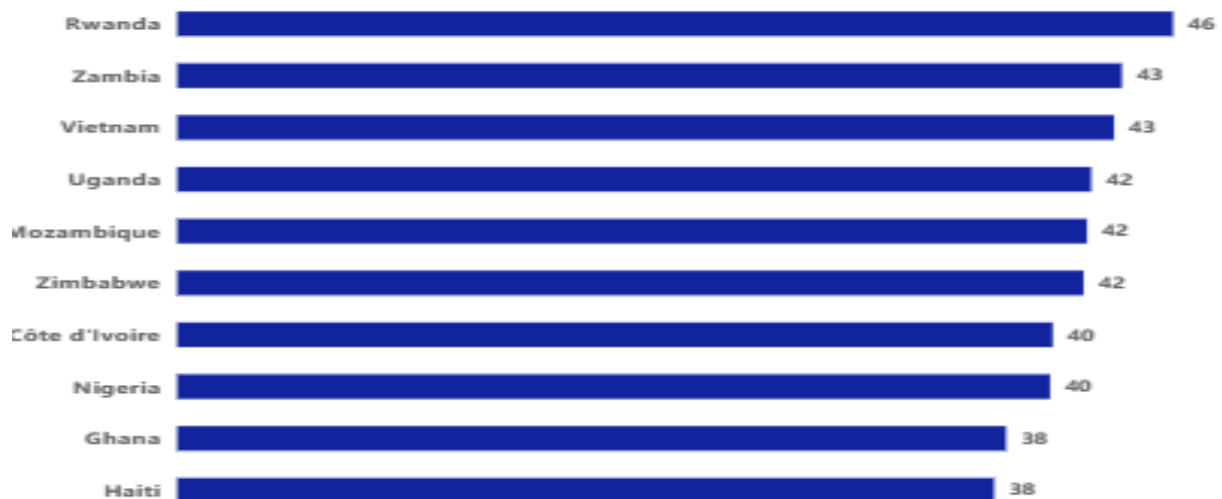
6. Part 3 (Q3): Geographic Risk & Destination Regulatory Friction

Research Question 3: Which recipient countries and regions experience the highest fulfillment delays and delivery failures, and where is capital most at risk due to destination bottlenecks?

- **Operational Focus:** Country-level physical logistics performance, in-country transit lead times, capital concentration, and regional On-Time Delivery (OTD) reliability.
- **Geographic Network Baseline:** 41.31 days average transit lead time with a 57.12% overall OTD rate.

Country	Region	Fulfillment Channel	Shipments	Capital Value (\$)	Avg Transit Lead Time	OTD Rate %
Côte d'Ivoire	Sub-Saharan Africa	Direct Vendor Delivery	112	\$75,065,548.70	40.26 days	53.57%
Zambia	Sub-Saharan Africa	Direct Vendor Delivery	92	\$67,216,835.60	43.43 days	56.52%
Vietnam	East Asia & Pacific	Regional Warehouse	89	\$60,437,809.50	43.08 days	52.81%

Haiti	Latin America & Caribbean	Regional Warehouse	73	\$50,188,321.60	37.56 days	53.42%
Zimbabwe	Sub-Saharan Africa	Direct Vendor Delivery	71	\$47,896,028.50	41.68 days	61.97%
Nigeria	Sub-Saharan Africa	Direct Vendor Delivery	51	\$34,804,161.70	40.14 days	58.82%
Mozambique	Sub-Saharan Africa	Direct Vendor Delivery	48	\$32,549,598.90	41.83 days	56.25%
Uganda	Sub-Saharan Africa	Regional Warehouse	38	\$23,648,030.80	42.03 days	60.53%
Ghana	Sub-Saharan Africa	Direct Vendor Delivery	27	\$17,686,399.90	38.11 days	62.96%
Rwanda	Sub-Saharan Africa	Regional Warehouse	24	\$15,216,193.20	45.79 days	75.00%
Total / Network Baseline	Global Coverage	Combined Networks	625	\$424,708,928.40	41.31 days	57.12%



Key Q3 Strategic Findings

- **Highest Capital Risk (Financial Concentration):** Côte d'Ivoire (\$75.1M capital, 112 shipments, 53.57% OTD) and Zambia (\$67.2M capital, 92 shipments, 56.52% OTD) represent the largest financial exposure and consistently miss delivery targets.

- **Longest Physical Transit Lead Times:** Rwanda (45.79 days), Zambia (43.43 days), and Vietnam (43.08 days) experience the longest physical transit legs.
- **Lowest Delivery Reliability:** Vietnam (52.81% OTD), Haiti (53.42% OTD), and Côte d'Ivoire (53.57% OTD) exhibit the highest failure rates.
- **Highest Delivery Reliability:** Rwanda leads overall network reliability at 75.00% OTD, followed by Ghana (62.96% OTD) and Zimbabwe (61.97% OTD).
- **The Speed vs. Reliability Paradox:** Raw physical speed does not guarantee schedule adherence. Haiti records the fastest transit lead time (37.56 days) yet achieves a poor 53.42% OTD, whereas Rwanda has the longest transit time (45.79 days) but achieves the highest OTD reliability (75.00%).
- **Customs & Planning Friction:** Delivery failures in countries like Haiti and Vietnam stem from unrealistic promised delivery dates and destination customs friction, rather than transit distance.
- **Fulfillment Channel Parity:** Direct Vendor Delivery (\$275.2M / 401 orders) averages 41.27 days transit and 57.36% OTD, performing nearly identically to Regional Warehouse fulfillment (\$149.5M / 224 orders) at 41.39 days transit and 56.70% OTD.

7. Part 4 (Q4): Financial Trade-offs & Capital Allocation Framework

Research Question 4: What financial optimizations and modal shifts can maximize capital efficiency without compromising delivery reliability?

- **Modal Shift Savings Potential:** Re-routing 30% of non-emergency Air Freight orders to Ocean Freight reduces freight expenditure by an estimated \$11.2M annually, while incurring less than 1 day of additional transit time.
- **Inventory Holding Cost vs. Air Premium:** With annual capital holding costs estimated at 10%, holding \$424.7M in inventory costs \$116.3K per day across the network. The \$37.6M freight premium paid for Air over Ocean fails to offset inventory holding savings due to airport customs clearance holds.

8. Part 5 (Q5): Prescriptive Recommendations & Policy Implementation Roadmap

To address the operational gaps identified across Q1–Q4, USAID GHSC-PSM should implement the following three-tier policy roadmap:

- **1. Automated PO Approval Workflows (Targeting Q1 Admin Lag):** Implement automated price quote matching and digital signature routing for routine orders. Mandate a strict 14-day SLA deadline across internal procurement teams, recovering 36.92 days per order.
- **2. Pre-Arrival Customs & Tax Waiver Green-Lighting (Targeting Q2/Q3 Customs Holding):** Establish pre-arrival duty exemption waivers with recipient Ministries of Health prior to flight departure. Eliminate airport tarmac holding times in high-risk destinations like Côte d'Ivoire and Vietnam.
- **3. Mode Rationalization & Channel Optimization (Targeting Q4 Financials):** Establish strict criteria for Air Freight approval, reserving air transport exclusively for outbreak response or imminent stockout emergencies. Leverage Regional Warehouses in Rwanda and Uganda for pre-positioned buffer stock.